

PADMAPRIYA RAJASEKARAN

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SUMMARY

Biomedical Engineering graduate and Lean Six Sigma Green Belt with hands-on experience in DOE, SPC, and cGMP manufacturing across FDA-regulated medical device environments

EDUCATION

B.S., Biomedical Engineering | Chemistry Minor

Rochester Institute of Technology, Rochester, NY

May 2026

Dean's List

SKILLS

Data Analysis: Design of Experiments (DOE), Statistical Process Control (SPC), Minitab, MATLAB, ANOVA / t-tests

Engineering: Process Validation (IQ/OQ/PQ), Yield & Cycle Time Optimization, Root Cause Analysis (RCA/CAPA), FMEA, Additive Manufacturing, Material Selection & Evaluation, cGMP, ISO 13485, SOP & Technical Writing

Software: SOLIDWORKS, Fusion 360, AutoCAD, LabVIEW, JIRA, Microsoft Office (Advanced Excel)

Certifications: Lean Six Sigma Green Belt (C.S.S.C)

WORK EXPERIENCE

Fresenius Medical Care, Lawrence, MA | NPI Systems Engineering Co-op

July 2025 – Dec 2025

- Tested a next-generation home hemodialysis system across 25+ operating conditions to confirm the device consistently delivered accurate fluid balance and conductivity readings within specification
- Analyzed 100,000+ pressure and flow data points per test run in MATLAB to identify why the system was producing measurement errors, pinpointing the root cause across 20+ trial datasets
- Executed parameter development test protocols and analyzed cyclor pressure and flow data to validate updated correction values, improving the system's volumetric accuracy
- Tracked system performance across 21+ verification trials using SPC to detect drift and confirm the device stayed consistent and repeatable over time
- Monitored 5 verification systems simultaneously, logging deviations in real time and recommending protocol improvements that eliminated unnecessary steps and improved test repeatability

Cranial Technologies, Chandler, AZ | R&D Engineering Co-op

July 2023 – May 2024

- Inspected 400+ custom 3D-printed infant cranial helmets weekly for surface defects and dimensional accuracy, catching and resolving production issues before they reached the next stage
- Developed optimal vapor-smoothing settings and build preparation parameters through structured DOE testing, improving surface finish and flexibility while minimizing material waste
- Designed and built custom jigs and cutting tools used on the production floor to trim and finish helmets consistently across different device configurations
- Authored 9 work instructions (117+ pages) covering every step of the production process so any team member could operate equipment and follow procedures the same way
- Built MATLAB scripts to automate helmet file preparation before printing, cutting processing time by 60%
- Compared three 3D printing materials through computational analysis to determine which material best balanced flexibility, durability, and fit for pediatric cranial orthoses
- Ran a head-to-head study comparing CNC-machined vs. 3D-printed helmet parts to determine which method produced better dimensional accuracy and was easier to manufacture

PROJECTS

Contact Lens Dry Processing | Bausch + Lomb, Rochester, NY

Jan 2026 – May 2026

- Conducted 120+ controlled experiments across 8 different mold surface treatments to figure out why contact lenses were sticking or deforming during release, systematically narrowing down the root cause
- Identified the key process variables driving defects such as surface preparation, temperature, and material interactions and optimized them to improve release yield by 19% and reduce production cycle time
- Inspected finished contact lenses for cosmetic defects after each trial run and calculated final release yield to validate whether process changes affected output quality